Can Blood Vessel Networks in Ultrafast Breast MRI Predict Breast Cancer Treatment Response?

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Why Might Blood Vessels Predict Treatment Response?

Neoadjuvant treatment (NAT) varies by breast cancer subtype and may include chemotherapy and HER2-targeted therapy. Only some patients achieve a **pathologic complete response (pCR)**, which is associated with improved long-term outcomes [1]. Earlier prediction could avoid ineffective treatment for women unlikely to benefit and identify nonresponders who need intensified or alternative therapy.

Tumors recruit abnormal vessels that can be dense, tortuous, leaky, and disorganized [3]. Their structure and contrast-enhancement dynamics may reflect tumor biology and affect drug delivery. We test whether these vessel-network features improve pCR prediction beyond simple summaries of the same MRI measurements.

Research Question: does the structure of breast blood vessel networks and the way contrast changes through them improve response prediction beyond simple summaries of the same MRI measurements?

UChicago Cohort: 179 Labeled, 1,226 Unlabeled Exams

UChicago contributed **181 labeled exams from 143 patients**; **179 exams from 141 patients** are analyzed here (115 non-pCR / **64 pCR**, 36%). Two exams lacked the required centerline, support mask, and aligned MRI.

Acquisition timing is heterogeneous: exams contain 13–24 frames spanning 96–480 s. Longitudinal exams are included. Cross-validation is grouped by patient, so one patient's exams never cross training/evaluation splits.

Pretraining draws on a separate, non-overlapping pool: **1,226 unlabeled exams from 225 patients** (1,081 CAPS, 145 CHIMEC), with no patient or exam shared with the labeled cohort.

Three Graph Representations Test Whether Topology Matters

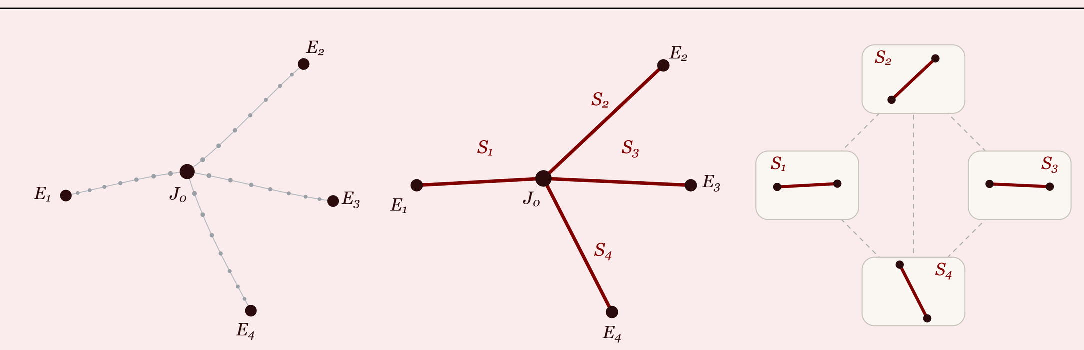

Figure 1. The same vessel centerline represented three ways.

We represent the vascular network of each patient's breasts as a graph:

- **Voxel graph:** each vessel-centerline voxel is a node; neighboring centerline voxels are connected.
- **Junction graph:** branch points are nodes ($v \in V_{\mathrm{vox}}, \deg(v) > 2$); vessel segments connect them.
- **Segment graph:** vessel segments are nodes; segments sharing an endpoint are connected.

Vessel Centerlines Become Graphs for pCR Prediction

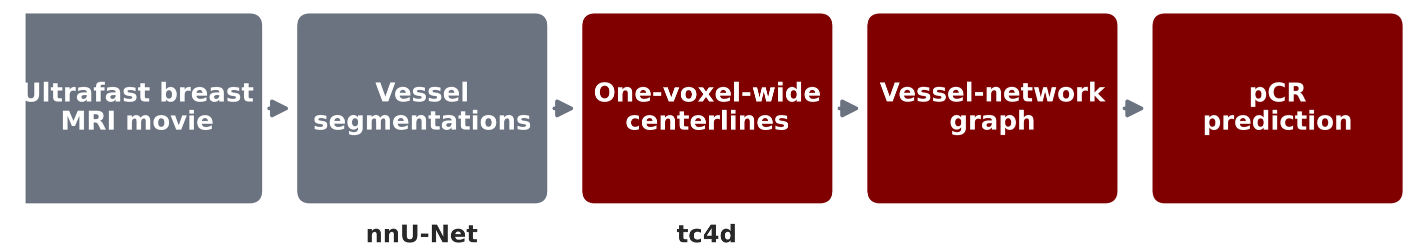

Figure 2. Ultrafast breast MRI movie to vessel-network prediction.

A published deep-learning method (nnU-Net) identifies vessel voxels [2]. Our **tc4d** algorithm converts that mask into a one-voxel-wide centerline while preserving vascular connectivity. The centerline then becomes one of the three graph representations.

Nodes carry geometry and measured enhancement curves. A **graph neural network (GNN)** combines neighboring-node information, pools the network into one exam representation, and predicts pCR.

Forecasting Pretraining Helps This GNN

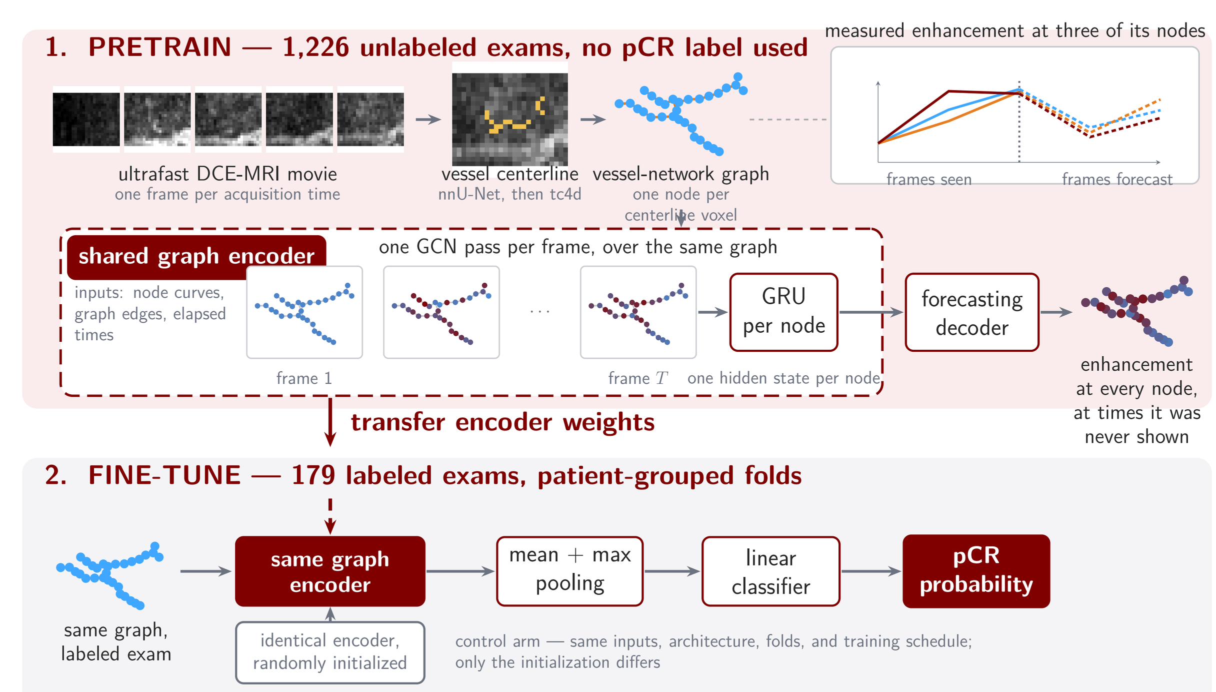

Figure 3. Forecasting weights transfer into the pCR classifier; a randomly initialized twin is the control arm.

The encoder receives vessel curves, connections, and elapsed times and predicts **future enhancement at connected vessel nodes** [4]. The downstream comparison holds inputs, encoder, training, patient-grouped folds, and evaluation fixed: forecasting versus random initialization, with and without graph message passing.

Centerline enhancement features predict pCR modestly, but hand-engineered GNNs do not beat matched tabular summaries. Forecasting pretraining improved this downstream GNN; graph-free pretraining was unresolved.

Clinical value remains unproven: this is one seed on one cohort. Repeated seeds and locked-cohort validation must reproduce the gain before any clinical or vascular-specific claim.

Forecast Pretraining Improves the GNN in One Development Run

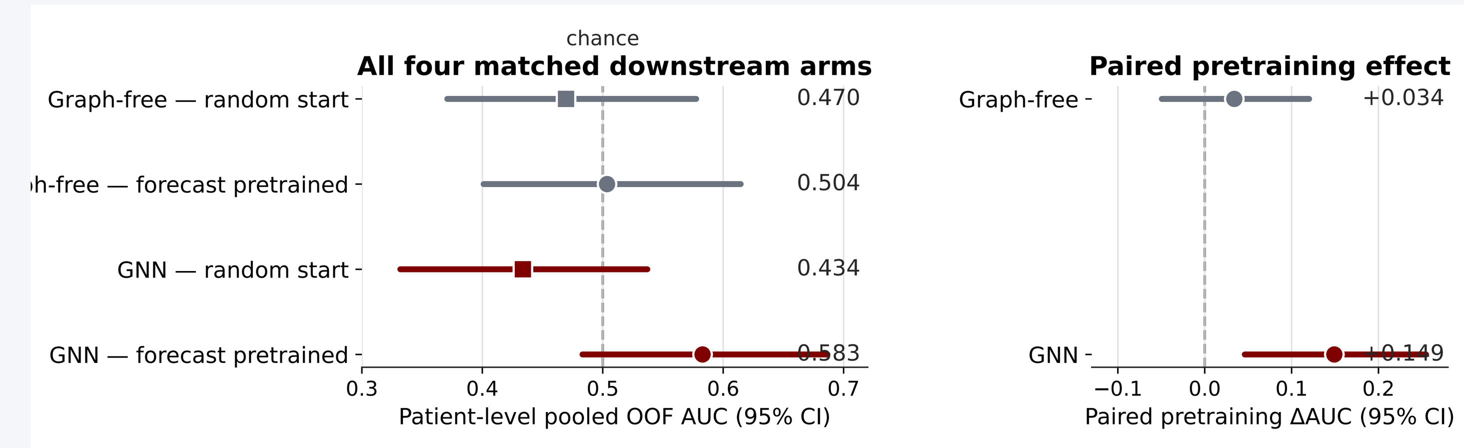

Figure 4. Patient-level pooled out-of-fold AUC with patient-bootstrap 95% CIs on shared patient-grouped folds.

Forecasting initialization changed GNN AUC from 0.434 to **0.583**: paired ΔAUC **+0.149** [**+0.046**, **+0.255**]. The graph-free effect was unresolved: +0.034 [−0.049, +0.120]. The pretrained GNN exceeded pretrained graph-free by +0.079 [+0.010, +0.155].

The claim is narrow: initialization helped this GNN on one seed. Its absolute interval includes chance. Earlier feature-matched experiments found that full kinetics drove most performance, with no measurable message-passing benefit.

Prediction Is Sensitive to Acquisition Timing

Acquisition timing is associated with outcome in this cohort: a model using frame count, duration, and cadence without imaging features reaches **0.587** AUC—essentially the pretrained GNN's 0.583. Their predictions correlate (Pearson $r = 0.50$).

This raises concern about protocol dependence, but cannot separate technical from biological association. Adjustment can remove genuine patient differences that happen to track acquisition as well as technical variation.

Acknowledgments + References

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